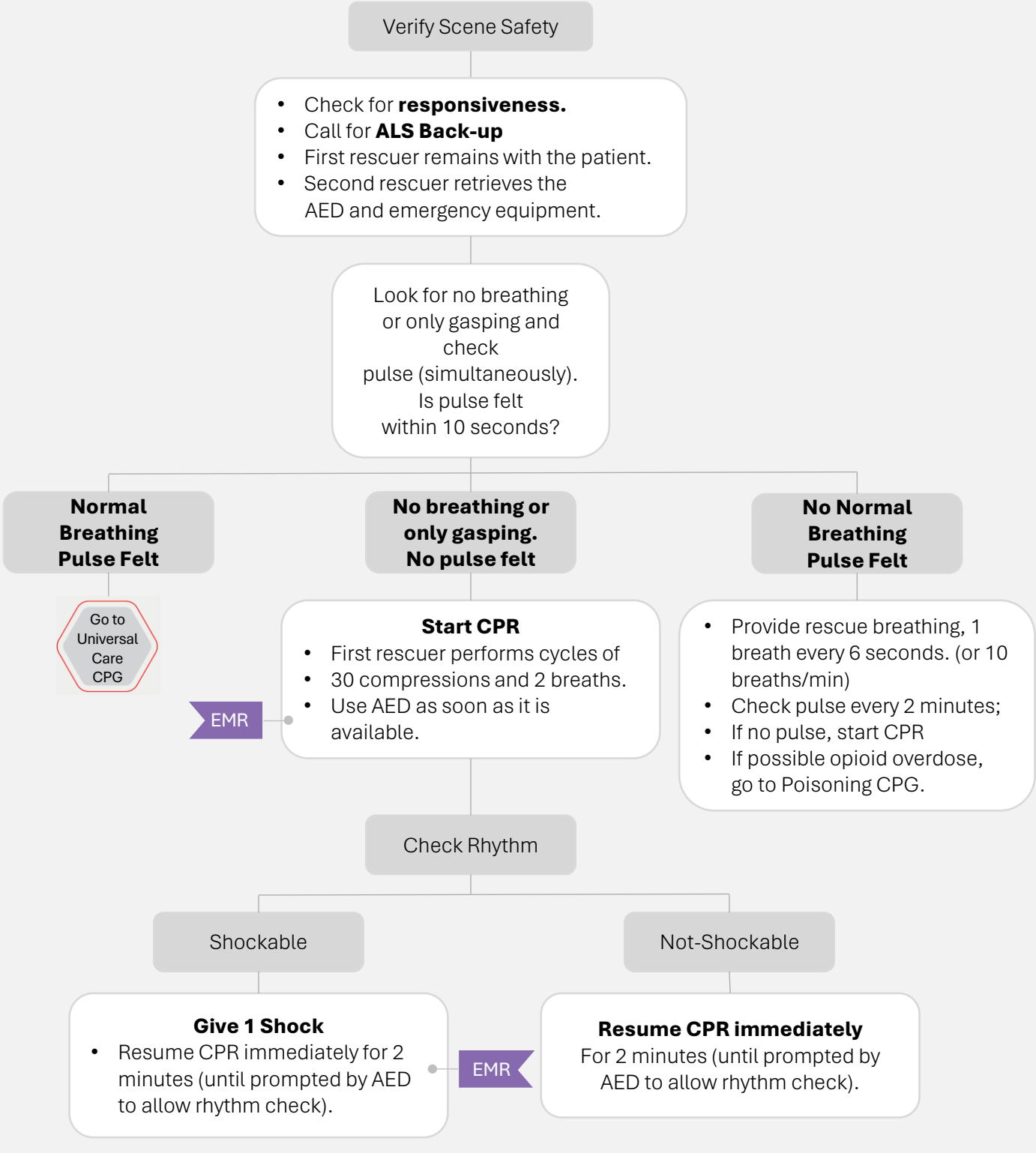
Adult Basic Life Support



Maternal Cardiac Arrest

Priorities for pregnant women in cardiac arrest include:

- Continuation of high-quality CPR with attention to good ventilation
- Left-lateral uterine displacement to relieve pressure on the major vessels in the abdomen to help with blood flow
- Transfer to an appropriate facility as soon as possible

Hypothermia

- Pulse check up to 60second

Patient Care Goals

1. Rapidly identify and initiate high-quality cardiopulmonary resuscitation (CPR) for adults in cardiac arrest.
2. Ensure ALS back-up is requested.
3. Rapidly apply AED (if available) and deliver a shock .

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Adult patients (≥13 years) presenting with cardiac arrest, including:
  - Unresponsiveness with no signs of life (e.g., no pulse, no breathing, or agonal gasps).
  - Suspected pulseless electrical activity (PEA), ventricular fibrillation (VF), ventricular tachycardia (VT), or asystole.
- Patients requiring immediate BLS interventions, including chest compressions, ventilations, and automated external defibrillator (AED) use.

Exclusion Criteria

- Patients with obvious signs of irreversible death (e.g., rigor mortis, dependent lividity, or decapitation; refer to Verification of Death CPG).
- Pediatric patients (<13 years).

Key Patient Assessment Points

1. **Cardiac Arrest Confirmation**
   - Verify unresponsiveness by assessing for no response to verbal or painful stimuli (AVPU).
   - Check for absence of pulse (carotid) for 5-10 seconds.
   - Confirm absence of normal breathing or presence of agonal gasps for 5-10 seconds.
2. **Scene and Bystander Assessment**
   - Ensure scene safety (e.g., no electrical or environmental hazards) before initiating BLS.
   - Determine if bystander CPR was initiated and its quality (e.g., rate, depth of compressions).
3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - All unconscious patients and patients in cardiac arrest are Red Priority.

Patient Management & Interventions

1. **Check for Responsiveness**
   - Verify unresponsiveness by assessing for no response to verbal or painful stimuli (AVPU).
2. **Call for ALS back-up**
3. **Retrieve Emergency Equipment**
   - First rescuer remains with the patient.
   - Second rescuer retrieves the AED and other emergency equipment.
4. **Primary Survey**
   - Perform Head Tilt/Chin lift (jaw thrust for suspected trauma patients)
   - Look for no breathing or only gasping and check pulse (simultaneously).
   - Is pulse felt within 10 seconds?
5. **No breathing or only gasping. No pulse felt**
   - Start CPR
   - First rescuer performs cycles of 30 compressions and 2 breaths.
   - Use AED as soon as it is available.
6. **No Normal Breathing. Pulse Felt**
   - Provide rescue breathing, 1 breath every 6 seconds. (or 10 breaths/min)
   - Check pulse every 2 minutes;
     - If no pulse, start CPR.
   - If possible opioid overdose, go to Overdose/Poisoning CPG.
7. **Apply AED**

- If shock is advised:
  - Give 1 Shock
  - Resume CPR immediately for 2 minutes (until prompted by AED to allow rhythm check).
- If no shock advised:
  - Resume CPR immediately
  - For 2 minutes (until prompted by AED to allow rhythm check).

8. **Continue CPR until ALS Back-up arrives**
9. **Check for signs of ROSC**
   - Check for signs of ROSC (e.g., spontaneous pulse, normal breathing) every 2 minutes.

Patient Safety Considerations

- None noted

Clinical Insights & Additional Information

- Chest Compressions
  - Initiate high-quality compressions immediately (100-120/min, 5-6 cm depth, full recoil).
  - Rotate compressors every 2 minutes to prevent fatigue and maintain quality.
  - Minimize interruptions (<10 seconds for rhythm checks or ventilations).
- Ventilations
  - Provide ventilations via bag-valve-mask (BVM) at 10-12 breaths/min (1 breath every 5-6 seconds) with adequate chest rise.
  - Use two-person BVM technique if available to improve seal and ventilation.
- Airway Management
  - Use oropharyngeal or nasopharyngeal airway if needed to maintain patency in unresponsive patients.
  - Suction secretions or vomit to prevent airway obstruction.
- Oxygen
  - Administer 100% oxygen via BVM to optimize oxygenation during CPR.

Documentation

- Identification: Patient's full name, gender, and age.
- Mechanism / Medical Complaint: Description of collapse or arrest, witness status, bystander CPR involvement.
- Injuries / Information: Initial ECG rhythm, suspected cause of arrest (e.g., MI, respiratory failure, trauma).
- Signs: Initial assessment findings, absence of breathing/pulse, ROSC indicators, capnography if available.
- Treatment:
  - Time CPR initiated, AED applied, and defibrillation administered.
  - Ventilation method and rate (e.g., BVM with OPA, 10-12 breaths/min).
  - Use of airway adjuncts, suctioning, oxygen administration.
- Allergies: Known allergies if information is available.
- Medications: Pre-arrest medications if known (e.g., insulin, beta blockers).
- Background: Any relevant past medical history from bystanders, family, or records.
- Other: Scene observations, handover time, ALS arrival, team member roles, witness details.
  - Timestamp all key actions: CPR start, defibrillation, ROSC.
  - Two full sets of vital signs post-ROSC (if applicable).
- ROSC documentation must include vitals, GCS, and interventions applied.

Key Performance Indicators

1. Initiate chest compressions within 1 minute of recognition of cardiac arrest.
2. Apply the AED within 3 minutes of recognition of cardiac arrest.

References

1. American Heart Association 2020, *2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care*, American Heart Association, Dallas